IPv6 asoslari

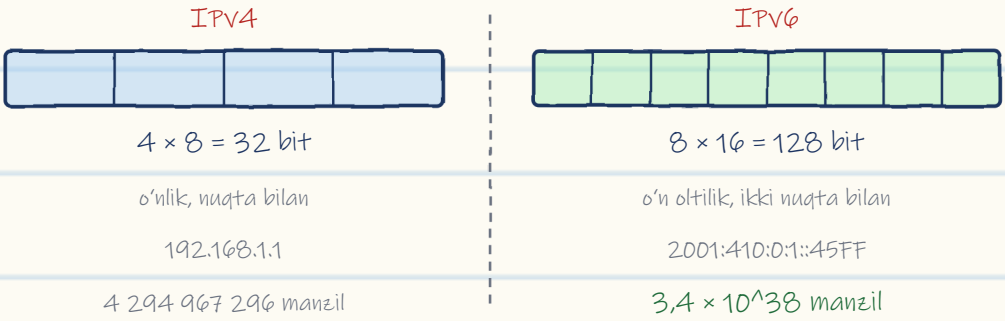
IPv6 Basics

Asosiy atamalar

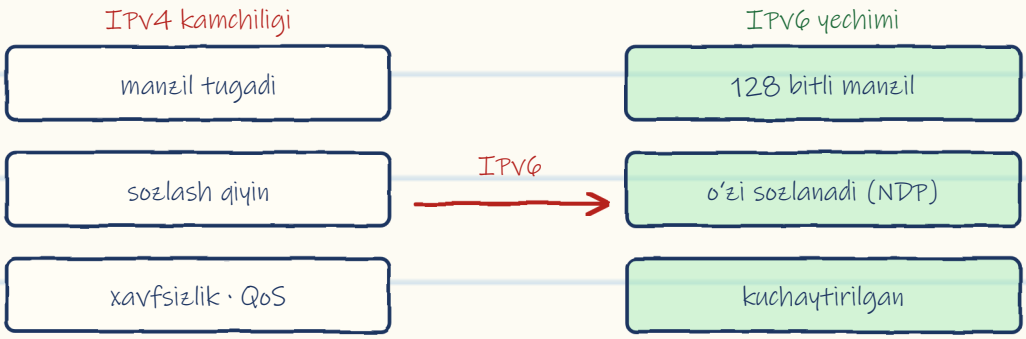
- IPv6 — 128 bitli manzil, o'n oltilik yozuvda
- Global unicast — Internetda yo'naltiriladigan yakka manzil
- Link-local — faqat bitta kanal ichida amal qiladi (FE80::/10)
- Anycast — bitta manzil, eng yaqin qurilma javob beradi
- NDP — qo'shnini aniqlash; IPv6 da ARP o'rnini bosadi

1. Nega IPv6 kerak?

- IPv4 — 32 bit, 1980-yillardan ishlaydi, manzil tugab bormoqda
- sozlash noqulay · xavfsizlik va QoS yetarli emas
- IPv6 (IPng) — IETF, 1990-yillardan; uzunlik 128 bit



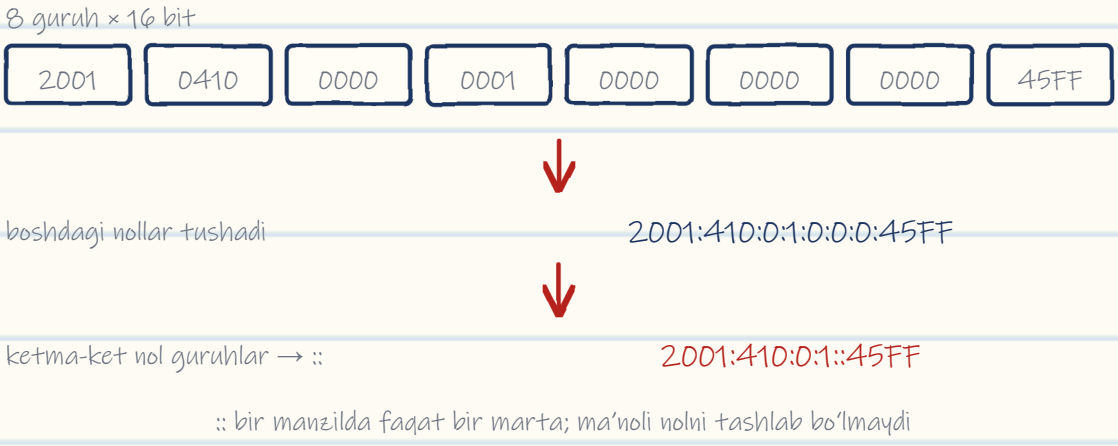
1-rasm. 32 bit → 128 bit: manzil fazosi to'rt barobar uzun



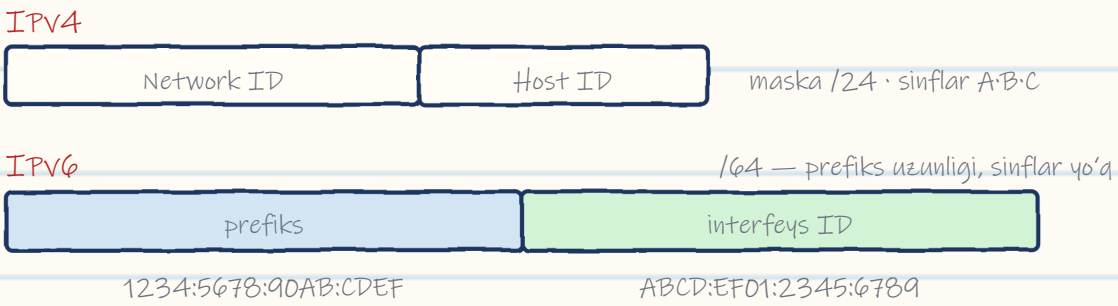
minimal sozlash · kuchaytirilgan xavfsizlik va QoS

2. Manzil yozilishi

8 guruh, har biri 16 bit — 4 ta o'n oltilik raqam, ikki nuqta bilan.



3. Manzil tarkibi

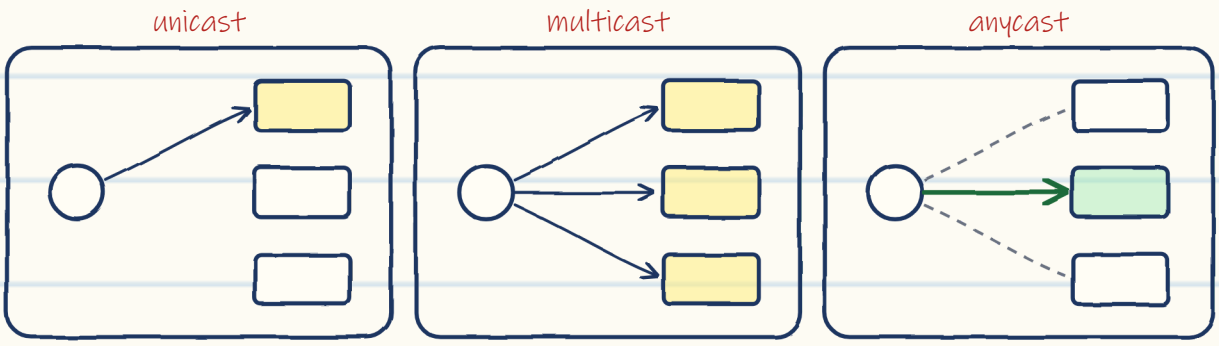


2-rasm. prefiks = tarmoq, interfeys ID = tugun (64 bit)

Mos kelishi: network ID → prefiks, host ID → interfeys ID, maska → prefiks uzunligi.

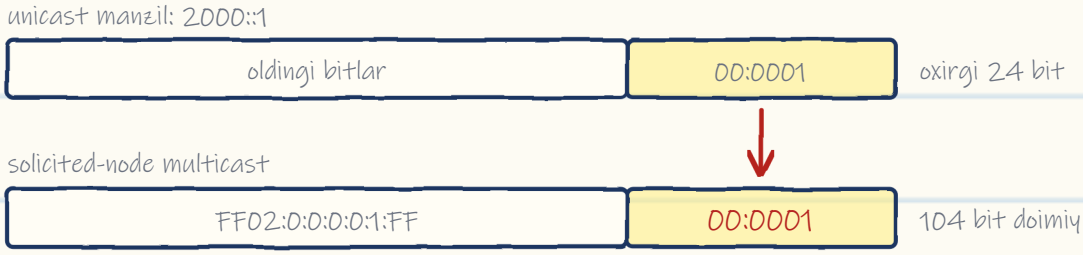
4. Manzil turlari

IPv6'da broadcast yo'q — uning o'rnini multicast bosadi.



anycast unicast fazosidan olinadi — yozilishi bilan farq qilmaydi

Turi	Prefiks	Izoh
belgilanmagan	::/128	faqat manba sifatida
loopback	::1/128	IPv4 127.0.0.1 kabi
link-local	FE80::/10	linkdan chiqmaydi
site-local	FEC0::/10	xususiy, kam ishlatiladi
global unicast	2000::/3	IANA beradi, ommaviy
multicast	FF00::/8	FF02::1 — hamma tugun, FF02::2 — hamma router

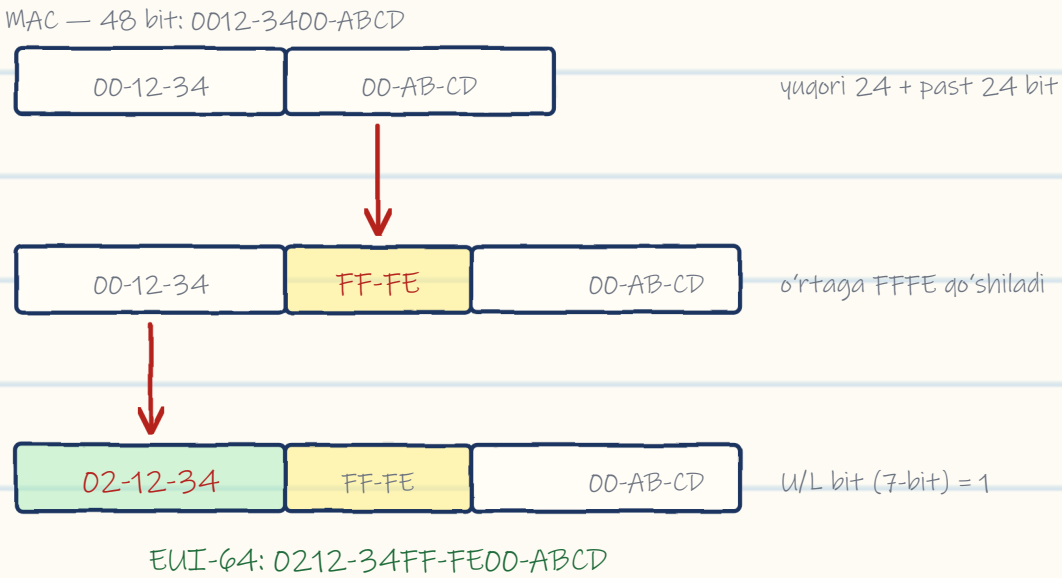


2000::1 → FF02::1:FF00:1

qo'shnining MAC manzilini so'rash va DAD uchun

3-rasm. Har bir unicast manzilning o'z solicited-node manzili bor

5. Interfeys ID — IEEE EUI-64



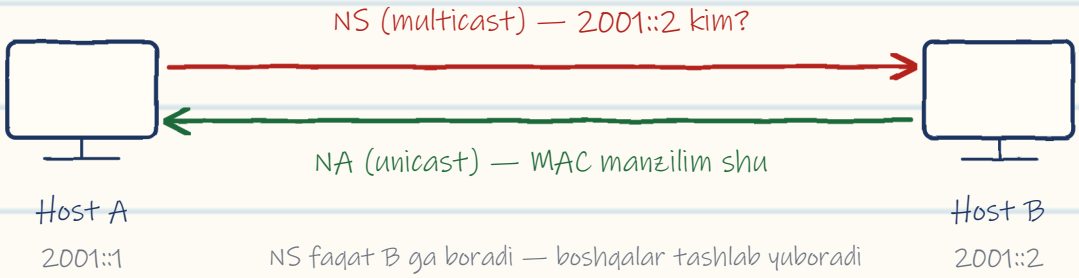
4-rasm. 64 bitli interfeys ID MAC manzildan olinadi

6. NDP vazifalari

manzil aniqlash	router/prefiks	avtosozlash	DAD
ARP o'rnida	prefiks va PMTU	manzilni o'zi yasaydi	takroriy manzilni topish

shu sabab IPV6'da DHCP shart emas

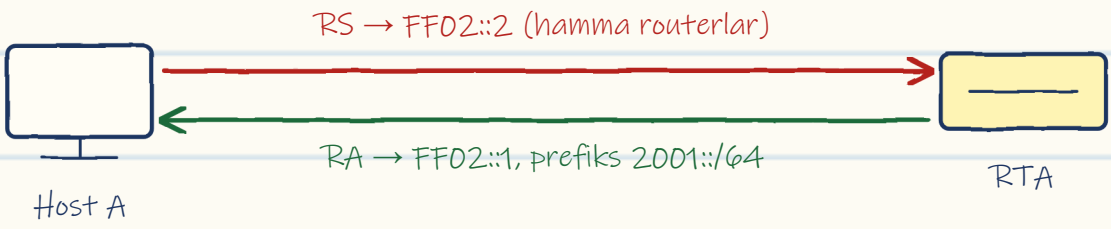
7. NS va NA — qo'shni MAC manzili



NS faqat B ga boradi — boshqalar tashlab yuboradi

javob unicast — multicast trafik kamayadi

8. Manzilni o‘zi sozlash





RA routerdan davriy ravishda ham keladi

5-rasm. RS/RA bilan prefiksni topish, so‘ng manzilni yig‘ish

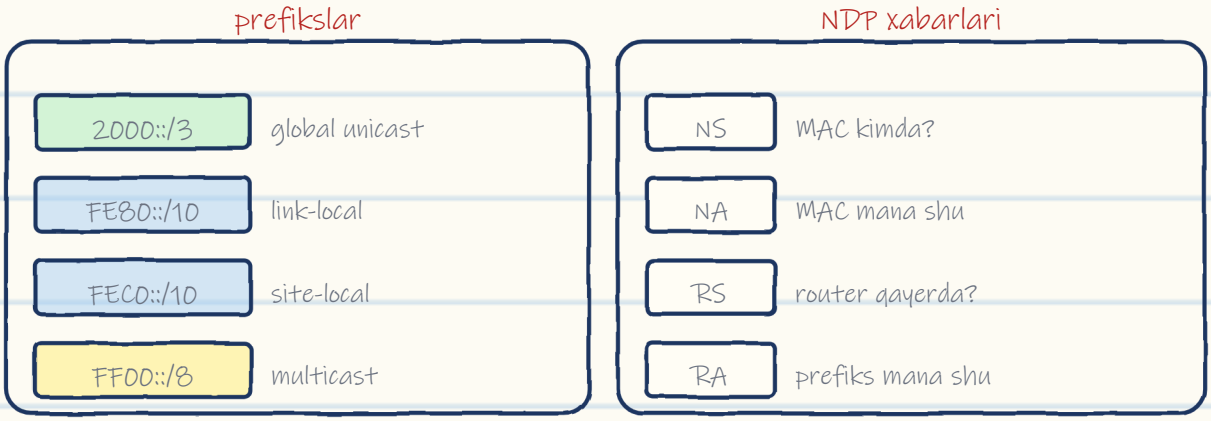
Buyruq (interfeys ko‘rinishida)	Nima qiladi
ipv6 address 2000::1/64	qo‘lda global unicast
ipv6 address auto	stateless avtomatik
ipv6 address FE80::1 link-local	qo‘lda link-local
ipv6 address auto link-local	avtomatik link-local

IPV6 sukut bo‘yicha yoqilgan; global manzil berilsa, link-local o‘zi paydo bo‘ladi.

```
display ipv6 interface GigabitEthernet 0/0
IPV6 is enabled, link-local address is FE80::1
Global unicast address(es): 2000::1, subnet is 2000::/64
Joined group address(es): FF02::1 · FF02::2 · FF02::1:FF00:1
MTU is 1500 bytes · ND DAD is enabled, attempts: 1
```

6-rasm. Tekshiruv: link-local, global va qo‘shilgan guruh manzillari

9. Xotira kartasi



Xulosa

- IPV6 — 128 bit, 8 guruh × 16 bit, o'n oltilik
- Qisqartirish: boshdagi nollar va bitta ::
- Prefiks + interfeys ID (64 bit, EUI-64)
- unicast · multicast · anycast; broadcast yo'q
- NDP: manzil aniqlash, router/prefiks topish, avtosozlash, DAD

Tekshirish

1. IPV6 uzunligi? — 128 bit
2. Link-local prefiksi? — FE80::/10
3. :: necha marta ishlatiladi? — bir marta
4. MAC o'rtasiga nima qo'shiladi? — FFFE
5. RS va RA manzillari? — FF02::2 va FF02::1